

Coupon Fit and R02 Release Record

Team and staff form | Complete the physical fit before the Final Wing run

Keep the R01 coupon and test it with the course-issued rod under staff supervision. Record the evidence below. A CAD hole or notebook message does not substitute for the physical fit test.

Team ID

R01 coupon ZIP filename

Coupon test date

Assigned printer and approved profile

Measured rod diameter in mm

Physical fit observations

Test the three dimple-marked holes without force or alteration. Their nominal radial clearances are 0.15, 0.25, and 0.35 mm.

Coupon marker	Radial clearance	Observed fit and defects
One dimple	0.15 mm	
Two dimples	0.25 mm	
Three dimples	0.35 mm	

Selected radial clearance in mm

Selected fit type snug or free slip

Why this hole was selected

Tester name

Staff supervisor

Evidence kept with the team

☐ Coupon photograph

☐ Rod measurement

☐ R01 ZIP

After this record is complete, set Final Wing, increment REVISION to R02, keep COUPON_REVISION as R01, and enter the measured fit data honestly.

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Manufacturing review | Complete before the October 13 printer appointment

The validated notebook ZIP supplies STEP and STL or 3MF geometry for each module. OrcaSlicer must still create printer-specific toolpaths using the staff-approved machine and material profile.

Final Wing ZIP filename and revision

STEP file present

Module 1 STL or 3MF filename

Module 2 STL or 3MF filename

Module 3 file or not used

CAD mass for full printed set in g

Geometry and notebook review

☐ Manual 450/160/100 mm geometry check signed

☐ Physical coupon record entered in R02

☐ Every module validated and bounded

☐ Full printed CAD mass at or below 300 g

☐ Assigned virtual-load FBD and root moment checked

OrcaSlicer and printer review

☐ Assigned K2 Pro and 0.4 mm nozzle selected

☐ Approved regular PLA Pro profile selected

☐ Imported as mm at 100 percent scale

☐ Slicer dimensions match CAD bounds

☐ First, middle, final, and all layers reviewed

☐ Shell, ribs, sleeves, holes, seams remain printable

Saved slicer project filename

Estimated print time and mass

Machine specific G-code filename

Generate G-code only from the approved OrcaSlicer profile. Do not edit G-code manually. Follow staff instructions for USB transfer and first-layer supervision.

Staff decision

☐ Release for assigned October 13 slot

☐ Hold and correct

Staff name and date

Assigned printer and slot

Corrections or conditions